Worked solutions to selected exercises from
Richard Hammack, *Book of Proof, edition 2.2*

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1 Sets

1.1 Introduction to Sets

A. Write each of the following sets by listing their elements between braces.

2. $\{3x+2 : x \in \mathbb{Z}\} \to \{..., -7, -4, -1, 2, 5, 8, 11, ...\}$
4. $\{x \in \mathbb{N} : -2 < x \leq 7\} \to \{1, 2, 3, 4, 5, 6, 7\}$
6. $\{x \in \mathbb{R} : x^2 = 9\} \to \{-3, 3\}$
8. $\{x \in \mathbb{R} : x^3 + 5x^2 = -6x\} \to \{-3, -2, 0\}$
10. $\{x \in \mathbb{R} : \cos x = 1\} \to \{2\pi x : x \in \mathbb{Z}\}$
12. $\{x \in \mathbb{Z} : |x| < 5\} \to \{-4, -3, -2, -1, 0, 1, 2, 3, 4\}$
14. $\{5x : x \in \mathbb{Z}, |2x| \leq 8\} \to \{-20, -15, -10, -5, 0, 5, 10, 15, 20\}$
16. $\{6a + 2b : a, b \in \mathbb{Z}\} \to \{2k : k \in \mathbb{Z}\} = \{..., -4, -2, 0, 2, 4, ...\}$

B. Write each of the following sets in set-builder notation.

18. $\{0, 4, 16, 36, 64, 100, ...\} \to \{4k^2 : k \in \mathbb{Z}\}$
20. $\{..., -8, -3, 2, 7, 12, 17, ...\} \to \{7 - 5x : x \in \mathbb{Z}\}$
22. $\{3, 6, 11, 18, 27, 38, ...\} \to \{n^2 + 2 : n \in \mathbb{N}\}$
24. $\{-4, -3, -2, -1, 0, 1, 2\} \to \{x \in \mathbb{Z} : -4 \leq x \leq 2\}$
26. $\{..., \frac{1}{27}, \frac{1}{9}, \frac{1}{3}, 1, 3, 9, 27, ...\} \to \{3^x : x \in \mathbb{Z}\}$
28. $\{..., -\frac{3}{2}, -\frac{3}{4}, 0, \frac{3}{4}, \frac{3}{2}, \frac{9}{4}, 3, \frac{15}{4}, \frac{9}{2}, ...\} \to \{\frac{3}{4}x - \frac{3}{4} : x \in \mathbb{Z}\}$

C. Find the following cardinalities.

30. $|\{\{1,4\},a,b,\{\{3,4\}\},\{\emptyset\}\}| = 5$

32. $|\{\{\{1,4\},a,b,\{\{3,4\}\},\emptyset\}\}| = 1$

34. $|\{x \in \mathbb{N} : |x| < 10\}| = 9$

36. $|\{x \in \mathbb{N} : x^2 < 10\}| = 3$

38. $|\{x \in \mathbb{N} : 5x \leq 20\}| = 4$

1.2 The Cartesian Product

A. Write out the indicated sets by listing their elements between braces.

2. Suppose $A = \{\pi, e, 0\}$ and $B = \{0, 1\}$.

   a) $A \times B = \{(\pi,0),(\pi,1),(e,0),(e,1),(0,0),(0,1)\}$

   b) $B \times A = \{(0,\pi),(0,e),(0,0),(1,\pi),(1,e),(1,0)\}$

   c) $A \times A = \{(\pi,\pi),(\pi,e),(\pi,0),(e,\pi),(e,e),(e,0),(0,\pi),(0,e),(0,0)\}$

   d) $B \times B = \{(0,0),(0,1),(1,0),(1,1)\}$

   e) $A \times \emptyset = \emptyset$

   f) $(A \times B) \times B = \{$
   $((\pi,0),0),((\pi,1),0),((e,0),0),((e,1),0),$
   $((0,0),0),((0,1),0),((\pi,0),1),((\pi,1),1),$
   $((e,0),1),((e,1),1),((0,0),1),((0,1),1)$
   $\}$

   g) $A \times (B \times B) = \{$
   $(\pi,(0,0)),(\pi,(0,1)),(\pi,(1,0)),(\pi,(1,1)),$
   $(e,(0,0)),(e,(0,1)),(e,(1,0)),(e,(1,1)),$
   $(0,(0,0)),(0,(0,1)),(0,(1,0)),(0,(1,1))$
   $\}$

   h) $A \times B \times B = \{$
   $(\pi,0,0),(\pi,0,1),(\pi,1,0),(\pi,1,1),$
   $(e,0,0),(e,0,1),(e,1,0),(e,1,1),$
   $(0,0,0),(0,0,1),(0,1,0),(0,1,1)$
   $\}$

4. $\{n \in \mathbb{Z} : 2 < n < 5\} \times \{n \in \mathbb{Z} : |n| = 5\} = \{(3,-5),(3,5),(4,-5),(4,5)\}$

6. $\{x \in \mathbb{R} : x^2 = x\} \times \{x \in \mathbb{N} : x^2 = x\} = \{(0,1),(1,1)\}$

8. $\{0,1\}^4 = \{$
$(0,0,0,0),(0,0,0,1),(0,0,1,0),(0,0,1,1),$
$(0,1,0,0),(0,1,0,1),(0,1,1,0),(0,1,1,1),$
$(1,0,0,0),(1,0,0,1),(1,0,1,0),(1,0,1,1),$
$(1,1,0,0),(1,1,0,1),(1,1,1,0),(1,1,1,1)$
$\}$

1.3 Subsets

A. List all the subsets of the following sets.

2. $\{1,2,\emptyset\} \to \{\},\{1\},\{2\},\{\emptyset\},\{1,2\},\{1,\emptyset\},\{2,\emptyset\},\{1,2,\emptyset\}$

4. $\emptyset \to \{\}$

6. $\{\mathbb{R},\mathbb{Q},\mathbb{N}\} \to \{\},\{\mathbb{R}\},\{\mathbb{Q}\},\{\mathbb{N}\},\{\mathbb{R},\mathbb{Q}\},\{\mathbb{R},\mathbb{N}\},\{\mathbb{Q},\mathbb{N}\},\{\mathbb{R},\mathbb{Q},\mathbb{N}\}$

8. $\{\{0,1\},\{0,1,\{2\}\},\{0\}\} \to \{\},\{\{0,1\}\},\{\{0,1,\{2\}\}\},\{\{0\}\},$
$\{\{0,1\},\{0,1,\{2\}\}\},\{\{0,1\},\{0\}\},\{\{0,1,\{2\}\},\{0\}\},\{\{0,1\},\{0,1,\{2\}\},\{0\}\}$

B. Write out the following sets by listing their elements between braces.

10. $\{X \subseteq \mathbb{N} : |X| \leq 1\} \to \{\emptyset,\{1\},\{2\},\{3\},...\}$

12. $\{X : X \subseteq \{3,2,a\}\ and\ |X| = 1\} \to \{3\},\{2\},\{a\}$

C. Decide if the following statements are true or false.

14. $\mathbb{R}^2 \subseteq \mathbb{R}^3 \to false$

16. $\{(x,y) : x^2 - x = 0\} \subseteq \{(x,y) : x - 1 = 0\} \to false$

1.4 Power Sets

A. Find the indicated sets.

2. $\mathcal{P}(\{1,2,3,4\}) = \{$
$\emptyset,\{1\},\{2\},\{3\},\{4\},$
$\{1,2\},\{1,3\},\{1,4\},\{2,3\},\{2,4\},\{3,4\},$
$\{1,2,3\},\{1,2,4\},\{1,3,4\},\{2,3,4\},$
$\{1,2,3,4\}$
$\}$

4. $\mathcal{P}(\{\mathbb{R},\mathbb{Q}\}) = \{\emptyset,\{\mathbb{R}\},\{\mathbb{Q}\},\{\mathbb{R},\mathbb{Q}\}\}$

6. $\mathcal{P}(\{1,2\}) \times \mathcal{P}(\{3\}) = \{$
$(\emptyset, \emptyset), (\emptyset, \{3\}),$
$(\{1\}, \emptyset), (\{1\}, \{3\}),$
$(\{2\}, \emptyset), (\{2\}, \{3\}),$
$(\{1,2\}, \emptyset), (\{1,2\}, \{3\})$
$\}$

8. $\mathcal{P}(\{1,2\} \times \{3\}) = \{\emptyset, \{(1,3)\}, \{(2,3)\}, \{(1,3), (2,3)\}\}$

10. $\{X \in \mathcal{P}(\{1,2,3\}) : |X| \leq 1\} = \{\emptyset, \{1\}, \{2\}, \{3\}\}$

12. $\{X \in \mathcal{P}(\{1,2,3\}) : 2 \in X\} = \{\{2\}, \{1,2\}, \{2,3\}, \{1,2,3\}\}$

B. Suppose that $|A| = m$ and $|B| = n$. Find the following cardinalities.

14. $|\mathcal{P}(\mathcal{P}(A))| = 2^{2^m}$

16. $|\mathcal{P}(A) \times \mathcal{P}(B)| = 2^{m+n}$

18. $|\mathcal{P}(A \times \mathcal{P}(B))| = 2^{m \cdot 2^n}$

20. $|\{X \subseteq \mathcal{P}(A) : |X| \leq 1\}| = 2^m + 1$

1.5 Union, Intersection, Difference

2. Suppose $A = \{0,2,4,6,8\}$, $B = \{1,3,5,7\}$ and $C = \{2,8,4\}$, Find:

a) $A \cup B = \{0,1,2,3,4,5,6,7,8\}$

b) $A \cap B = \emptyset$

c) $A - B = A$

d) $A - C = \{0,6\}$

e) $B - A = B$

f) $A \cap C = C$

g) $B \cap C = \emptyset$

h) $B \cup C = \{1,2,3,4,5,7,8\}$

i) $C - B = C$

4. Suppose $A = \{b,c,d\}$ and $B = \{a,b\}$. Find:

a) $(A \times B) \cap (B \times B) = \{(b,a), (b,b)\}$

b) $(A \times B) \cup (B \times B) = \{(b,a), (b,b), (c,a), (c,b), (d,a), (d,b), (a,a), (a,b)\}$

c) $(A \times B) - (B \times B) = \{(c, a), (c, b), (d, a), (d, b)\}$

d) $(A \cap B) \times A = \{(b, b), (b, c), (b, d)\}$

e) $(A \times B) \cap B = \emptyset$

f) $\mathcal{P}(A) \cap \mathcal{P}(B) = \{\emptyset, \{b\}\}$

g) $\mathcal{P}(A) - \mathcal{P}(B) = \{\{c\}, \{d\}, \{b, c\}, \{b, d\}, \{c, d\}, \{b, c, d\}\}$

h) $\mathcal{P}(A \cap B) = \{\emptyset, \{b\}\}$

i) $\mathcal{P}(A) \times \mathcal{P}(B) = \{$
$(\emptyset, \emptyset), (\emptyset, \{a\}), (\emptyset, \{b\}), (\emptyset, \{a, b\}),$
$(\{b\}, \emptyset), (\{b\}, \{a\}), (\{b\}, \{b\}), (\{b\}, \{a, b\}),$
$(\{c\}, \emptyset), (\{c\}, \{a\}), (\{c\}, \{b\}), (\{c\}, \{a, b\}),$
$(\{d\}, \emptyset), (\{d\}, \{a\}), (\{d\}, \{b\}), (\{d\}, \{a, b\}),$
$(\{b, c\}, \emptyset), (\{b, c\}, \{a\}), (\{b, c\}, \{b\}), (\{b, c\}, \{a, b\}),$
$(\{b, d\}, \emptyset), (\{b, d\}, \{a\}), (\{b, d\}, \{b\}), (\{b, d\}, \{a, b\}),$
$(\{c, d\}, \emptyset), (\{c, d\}, \{a\}), (\{c, d\}, \{b\}), (\{c, d\}, \{a, b\}),$
$(\{b, c, d\}, \emptyset), (\{b, c, d\}, \{a\}), (\{b, c, d\}, \{b\}), (\{b, c, d\}, \{a, b\})$
$\}$

10. Do you think the statement $(\mathbb{R} - \mathbb{Z}) \times \mathbb{N} = (\mathbb{R} \times \mathbb{N}) - (\mathbb{Z} \times \mathbb{N})$ is true, or false? Justify.

True. An ordered pair (x, n) lies in $(\mathbb{R} \times \mathbb{N}) - (\mathbb{Z} \times \mathbb{N})$ exactly when $(x, n) \in (\mathbb{R} \times \mathbb{N})$ but $(x, n) \notin (\mathbb{Z} \times \mathbb{N})$, that is, $x \in \mathbb{R}$ and $n \in \mathbb{N}$ but not both $x \in \mathbb{Z}$ and $n \in \mathbb{N}$. Since in both cases $n \in \mathbb{N}$, the second premise collapses to $x \notin \mathbb{Z}$, hence, the ordered pair satisfies $x \in (\mathbb{R} - \mathbb{Z})$ and $n \in \mathbb{N}$, that is, $(x, n) \in (\mathbb{R} - \mathbb{Z}) \times \mathbb{N}$, therefore, both sets have exactly the same elements.

1.6 Complement

2. Let $A = \{0, 2, 4, 6, 8\}$ and $B = \{1, 3, 5, 7\}$ have universal set $U = \{0, 1, 2, ..., 8\}$. Find:

a) $\bar{A} = B$

b) $\bar{B} = A$

c) $A \cap \bar{A} = \emptyset$

d) $A \cup \bar{A} = U$

e) $A - \bar{A} = A$

f) $\overline{A \cup B} = \overline{U} = \emptyset$

g) $\bar{A} \cap \bar{B} = \emptyset$

h) $\overline{A \cap B} = U$

i) $\bar{A} \times B = B \times B = \{$
$(1,1),(1,3),(1,5),(1,7),$
$(3,1),(3,3),(3,5),(3,7),$
$(5,1),(5,3),(5,5),(5,7),$
$(7,1),(7,3),(7,5),(7,7)$
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1.7 Venn Diagrams

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1.8 Indexed Sets

2. Suppose $A_1 = \{0,2,4,6,8,10,12,14,16,18,20,22,24\}$**,** $A_2 = \{0,3,6,9,12,15,18,21,24\}$**,** $A_3 = \{0,4,8,12,16,20,24\}$**.**

a) $\bigcup_{i=1}^{3} A_i = \{0,2,3,4,6,8,9,10,12,14,15,16,18,20,21,22,24\}$

b) $\bigcap_{i=1}^{3} A_i = \{0,12,24\}$

4. For each $n \in \mathbb{N}$**, let** $A_n = \{-2n, 0, 2n\}$**.**

a) $\bigcup_{i \in \mathbb{N}} A_i = \{2n : n \in \mathbb{Z}\}$

b) $\bigcap_{i \in \mathbb{N}} A_i = \{0\}$

6.

a) $\bigcup_{i \in \mathbb{N}} [0, i+1] = [0, \infty)$

b) $\bigcap_{i \in \mathbb{N}} [0, i+1] = [0, 2]$

8.

a) $\bigcup_{\alpha \in \mathbb{R}} \{\alpha\} \times [0,1] = \mathbb{R} \times [0,1] = \{(x,y) : x \in \mathbb{R},\ 0 \leq y \leq 1\}$

b) $\bigcap_{\alpha \in \mathbb{R}} \{\alpha\} \times [0,1] = \emptyset$

10.

a) $\displaystyle\bigcup_{x\in[0,1]} [x,1]\times[0,x^2] = \{(u,v) : 0 \le u \le 1,\ 0 \le v \le u^2\}$

b) $\displaystyle\bigcap_{x\in[0,1]} [x,1]\times[0,x^2] = \{(1,0)\}$

12. If $\displaystyle\bigcap_{\alpha\in I} A_\alpha = \bigcup_{\alpha\in I} A_\alpha$, what do you think can be said about the relationships between the sets A_α?

They must all be equal.

14. If $J \neq \emptyset$ and $J \subseteq I$, does it follow that $\displaystyle\bigcap_{\alpha\in I} A_\alpha \subseteq \bigcap_{\alpha\in J} A_\alpha$? Explain.

Yes.
Assume $J \neq \emptyset$ and $J \subseteq I$, then,
take $x \in \bigcap_{\alpha\in I} A_\alpha$, this means $x \in A_\alpha$ for every $\alpha \in I$,
since $J \subseteq I$, then every element of J is also an element of I,
thus $x \in A_\alpha$ for every $\alpha \in J$, then, by definition, $x \in \bigcap_{\alpha\in J} A_\alpha$,
so we conclude that $\displaystyle\bigcap_{\alpha\in I} A_\alpha \subseteq \bigcap_{\alpha\in J} A_\alpha$. $\square$